Muhammad Umar Khan

Email: m.umark2002@gmail.com | (226) 506-9244 | LinkedIn| GitHub | Portfolio

SUMMARY

M.S. Computer Science student with hands-on experience building full-stack platforms and AI-driven systems from development to cloud deployment. Designed async processing pipelines, LLM-integrated APIs, and JWT-secured backends across Java, C#, and Python stacks including Spring Boot, ASP.NET Core, React, PostgreSQL, and AWS.

EDUCATION

Wayne State University | Detroit, MI, USA **Jan. 2025 - Aug. 2026**
M.S. in Computer Science |GPA: 3.92

University of Windsor | Windsor, ON, Canada **Sep. 2020 – Apr. 2024**
B.S. in Computer Science with Software Engineering Specialization

TECHNICAL SKILLS

Languages: Java, Python, C#, JavaScript/TypeScript, SQL
Frameworks: Spring Boot, Spring Security, ASP.NET, MVC, .NET 10, FastAPI, Flask, Node.js, React, Next.js, Razor
Databases: PostgreSQL, MongoDB, SQLite
Tools & DevOps: Git, Docker, AWS (Elastic Beanstalk, RDS, S3, CloudFront), REST APIs, JWT Authentication, Junit, Redis, Celery, Railway, Vercel, Gemini API
Frontend: HTML, CSS, Tailwind CSS, Vite

EXPERIENCE

University of Windsor| Windsor, ON, Canada **Apr. 2023 - Nov. 2024**
Research Assistant

- Engineered an image segmentation pipeline from scratch using KMeans clustering and Euclidean distance calculations to partition images into superpixel regions with configurable overlap.
- Studied and reproduced hybrid CNN+GNN architecture from published research, implementing SLIC-based superpixel graph construction and a coupled model for image classification
- Implemented and benchmarked CNN (TensorFlow/sklearn) and GCN (PyTorch Geometric) models, achieving 98% on MNIST and 81% on Cora, comparing classification accuracy across architectures.

University of Windsor| Windsor, ON, Canada **Sep. 2022 – Apr. 2024**
Teaching Assistant

- Mentored 100+ students through code reviews and debugging sessions across data structures, OOP, and databases.
- Led weekly lab sessions on algorithmic complexity and query optimization, helping students write cleaner, more efficient code.

PROJECTS

AutoOps AI | *ASP.NET Core, C#, React, TypeScript, FastAPI, PostgreSQL, Redis, Celery, Docker, Railway, Vercel*

- Built a full-stack AI document processing platform using ASP.NET Core, JWT auth with refresh token flow, and EF Core over PostgreSQL.
- Designed a Redis-backed Celery worker pipeline processing documents through parse → chunk → analyze stages with real-time SSE progress updates streamed to the React frontend
- Containerized with Docker Compose and deployed to Railway and Vercel with environment-driven config (CORS, JWT, Redis).

Khan's Blog Platform | *Java, Spring Boot, React, TypeScript, PostgreSQL, AWS*

- Deployed a full-stack blog platform on AWS using Elastic Beanstalk (backend), RDS PostgreSQL (database), S3 and CloudFront (frontend CDN), with all API traffic proxied through CloudFront for unified HTTPS delivery.
- Implemented JWT-based authentication with RBAC restricting admin/author permissions and built a rich text editor using TipTap with DOMPurify sanitization.

AdaptIQ | *React, FastAPI, Python, JavaScript, Google Gemini API, Railway*

- Built AI resume optimizer using a single-pass LLM pipeline for keyword extraction, cutting API overhead by 50%.
- Deployed to Railway with JWT middleware protecting all API routes and React served as static files from FastAPI
- Designed a 5-metric ATS scoring engine (parsing ability, keyword injection, coverage, quantification rate, job match) that programmatically scans tailored resume text and surfaces actionable improvement insights per generation.